

MAHABAD, Iran, Islamic Republic of

WMO: 407260

Lat: 36.7529N

Lon: 45.7153E

Elev: 4436

StdP: 12.49

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	16.1	3.5	7.8	16.7	7.4	9.5	22.7	34.0	41.0	30.8	41.0	0.0	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	98.3	61.1	95.4	60.7	92.8	60.7	65.4	89.2	64.3	87.7	63.1	86.4	11.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.5	80.2	76.7	55.0	75.9	76.4	53.6	72.0	75.7	32.8	89.1	31.8	87.9	30.9	86.3	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.0	22.9	20.0	5.8	103.4	5.5	1.9	1.9	104.8	-1.3	105.9	-4.4	107.0	-8.4	108.4		
(5)			3.6	67.3	4.9	1.9	0.1	68.7	-2.8	69.8	-5.5	70.8	-9.1	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.9	32.4	37.5	46.8	54.7	64.6	74.1	79.9	79.1	71.1	59.4	45.8	36.2		
	DBStd	17.58	8.50	7.49	7.47	6.16	4.84	4.15	4.11	3.82	4.76	6.66	7.15	8.78		
	HDD50	1672	545	349	150	26	0	0	0	0	0	9	161	432		
	HDD65	4395	1010	769	564	313	69	1	0	0	8	193	575	893		
	CDD50	4200	0	1	52	165	453	724	928	901	634	302	36	4		
	CDD65	1448	0	0	1	3	57	275	463	436	192	21	0	0		
	CDH74	17412	0	0	3	50	586	3082	5704	5512	2242	233	0	0		
	CDH80	8292	0	0	0	5	103	1302	3040	2951	862	29	0	0		
	WSAvg	7.2	7.0	7.6	10.0	8.2	8.5	7.5	6.9	6.1	6.2	6.5	5.6	6.0		
	PrecAvg	14.4	1.8	1.5	2.2	2.5	1.3	0.2	0.1	0.0	0.1	1.0	1.9	1.7		
(15) Precipitation	PrecMax	24.1	3.1	4.7	4.0	5.2	4.0	1.5	0.5	0.4	0.8	4.8	5.3	4.4		
	PrecMin	8.9	0.6	0.1	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1		
	PrecStd	3.7	0.7	1.0	0.9	1.3	1.0	0.3	0.2	0.1	0.2	1.3	1.4	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.6	57.7	73.3	79.5	86.3	96.0	102.7	101.3	95.0	83.4	69.3	60.0		
		MCWB	40.0	42.5	49.8	55.6	57.4	59.5	61.8	61.5	59.8	55.6	49.4	44.9		
	2%	DB	48.9	54.1	68.2	74.4	82.8	92.6	99.2	98.0	90.9	79.8	64.3	55.0		
		MCWB	39.1	42.1	48.1	53.3	57.1	59.4	61.4	61.0	58.6	54.2	47.8	42.2		
	5%	DB	46.1	51.0	63.1	70.8	80.2	90.0	96.1	95.4	87.9	76.5	60.6	51.0		
		MCWB	38.2	40.8	46.4	52.3	56.6	58.6	61.1	60.8	58.0	53.5	46.5	41.0		
	10%	DB	43.7	48.4	59.3	67.3	77.3	87.3	93.0	92.9	85.1	72.8	57.4	47.7		
		MCWB	36.7	39.7	45.0	50.9	55.8	58.4	61.1	61.1	57.2	52.8	46.0	39.5		
	0.4%	WB	43.5	45.8	51.1	57.0	62.8	64.4	66.9	67.1	63.4	58.6	52.8	46.7		
		MCDB	48.5	53.9	67.5	73.3	79.0	87.9	91.5	90.5	86.5	76.9	62.2	55.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.4	43.3	49.1	55.2	59.8	62.5	65.6	65.4	61.2	56.6	50.7	43.9		
		MCDB	46.9	51.1	64.1	70.9	76.5	85.9	89.6	89.2	84.3	73.4	58.8	52.2		
	5%	WB	38.6	41.5	47.4	53.7	58.2	61.4	64.6	64.3	59.8	55.2	49.0	41.8		
		MCDB	44.9	49.5	61.8	67.0	75.3	84.2	88.0	88.1	82.6	71.1	56.8	49.4		
	10%	WB	37.1	39.9	45.6	52.2	56.9	60.2	63.4	63.2	58.7	53.9	47.0	40.1		
		MCDB	42.9	47.4	58.2	64.6	73.9	82.3	86.7	86.8	80.9	68.9	55.3	46.8		
	5% DB	MDBR	13.1	15.1	17.2	19.6	21.9	24.6	24.3	26.7	26.7	21.7	17.0	12.9		
		MCDBR	15.2	19.0	22.5	25.5	25.8	28.5	28.8	30.7	30.8	26.8	22.0	16.9		
		MCWBR	8.9	10.2	9.9	10.8	9.8	9.9	9.7	10.7	11.8	10.9	11.0	9.4		
		MCWBR	13.2	16.8	19.4	21.6	24.1	25.3	24.1	26.1	27.6	22.9	16.5	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.339	0.387	0.454	0.450	0.407	0.436	0.405	0.362	0.381	0.327	0.305		
		taud	2.349	2.263	2.137	1.972	2.019	2.174	2.095	2.183	2.297	2.199	2.356	2.388		
	Ebn at Noon		279	281	277	263	266	277	268	274	282	262	268	272		
		Edh at Noon	30	36	45	57	55	47	50	45	38	38	29	27		
(44) All-Sky Solar Radiation	RadAvg		794	1089	1383	1731	2164	2619	2519	2309	1912	1307	884	688		
	RadStd		75	110	130	151	131	109	59	74	100	100	116	91		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	-3.24	N/A	+1.87	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page